

Introduction to Explainable AI

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ABSTRACT

As Artificial Intelligence technologies are increasingly used to make important decisions and perform autonomous tasks, providing explanations to allow users and stakeholders to understand the AI has become a ubiquitous concern. Recently, a number of open-source toolkits are making the growing collection of Explainable AI (XAI) techniques accessible for researchers and practitioners to incorporate explanation features in AI systems. This course is open to anyone interested in implementing, designing and researching on the topic, aiming to provide an overview on the technical and design methods for XAI, as well as hands-on experience with an XAI toolkit.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

AI; machine learning; explainable AI; interpretable machine learning; human-AI interaction

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1 INTRODUCTION

Artificial Intelligence (AI) technologies are increasingly used to make decisions and perform autonomous tasks in critical domains such as healthcare, finance, and employment. The needs to understand AI in order to improve, contest, develop appropriate trust and better interact with AI systems have spurred great academic and public interest in Explainable AI (XAI). On the one hand, the rapidly growing collection of XAI techniques allows diverse styles

of explanations to be incorporated in AI systems. On the other hand, to deliver satisfying user experiences with AI explanations requires user-centered approaches and interdisciplinary research to connect user needs and technical advancement. In short, XAI is an area with growing needs and exciting opportunities for HCI research.

This course is intended for HCI researchers and practitioners who are interested in developing and designing explanation features in AI systems, and those who want to understand the trends and core topics in the XAI literature. The course will introduce available toolkits that make it easy to create explanations for ML models, including AIX 360 [1] (<http://aix360.mybluemix.net>), a comprehensive toolkit providing technical and educational resources on the topic such as introduction to XAI concepts, python code libraries, and tutorials. We will also draw on our experience working with industry design practitioners to discuss opportunities and challenges to incorporate state-of-the-art XAI techniques in AI systems, including a “question-driven XAI design process” we developed in the our research [7].

The course will address the following questions:

- *What is XAI?* What problems does XAI solve? What are the focuses of XAI work in different research communities (e.g., Machine Learning, HCI)?
- *Why is XAI important?* What are the motivations for XAI? What applications or situations need XAI?
- *How to explain?* What are the state-of-the-art XAI techniques? How to determine their suitability for different AI applications, users and contexts? How to design XAI user experiences?
- *Where to start with XAI?* What tools are available for implementing XAI? What guidelines are available for designing XAI? Where to find relevant resources?

2 BENEFITS

Attendees of the course will benefit from:

- Having an overview of the concepts, trends and core topics in the scientific work of XAI.
- Gaining an understanding on common methods to explain ML models, and how to make appropriate choices of explanation methods and design explanation features that satisfy user needs.
- Learning about the availability of open-source toolkits and design guidelines for XAI.

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- Hands-on experience in creating different styles of explanation for a real-world use case of a ML system.

3 INTENDED AUDIENCE AND PREREQUISITES

The intended audience for this course are any CHI attendees who have already, or intend to engage in developing, designing and researching on the topic of XAI. The course does not require advanced knowledge in AI, data science or programming, though a basic understanding of machine learning concepts such as classification, training data, and features could be helpful. The course will include 15-20 minutes hands-on practice with Python code samples provided. Interested attendees could further explore the code but programming is not required. The course instructors will provide instructions to use the code samples, as well as introductory materials for machine learning and Python programming beforehand for interested attendees.

4 CONTENT

The course will be divided into four parts:

4.1 Part 1: Overview of XAI (20 minutes)

The course will begin with the context and motivation for the rapidly growing field of explainable AI, with examples use cases where explainability is critical, such as AI-assisted high-stakes decisions, supporting model development, ensuring safety and control of AI, as well as the requirement to address ethical and legal concerns. We will discuss core topics in the XAI literature [2, 6], including different types of explanation and evaluation of explainability. We will also discuss recent trends in HCI research towards user-centered XAI (e.g., [3, 4, 7, 8]).

4.2 Part 2: Introduction to XAI methods and toolkits (20 minutes)

We will then dive into 1-2 use cases of ML system to showcase a few popular explanation methods, and to demonstrate the diverse types of user needs for XAI. For example, we will introduce FICO Explainable Machine Learning Challenge [5], where a ML system is used to help loan officers decide whether a loan applicant qualifies for a line of credit. We will highlight the existence of different explanation consumers, i.e. a loan officer, a developer of the ML model, a loan applicant, and their different needs for understanding the ML system. With this use case, several popular XAI methods will be presented, such as explaining by decision rules the model follows, by how the model weighs features of a specific instance to arrive at a decision, and by similar examples.

This part will also cover an introduction to open-source toolkits that support creating explanations for ML models. In particular, we will discuss AIX360, a toolkit that provides technical and educational resources for XAI. It presents a taxonomy organizing the space of explainability methods¹ (Figure 1) together with Python code libraries and tutorials to implement these methods.

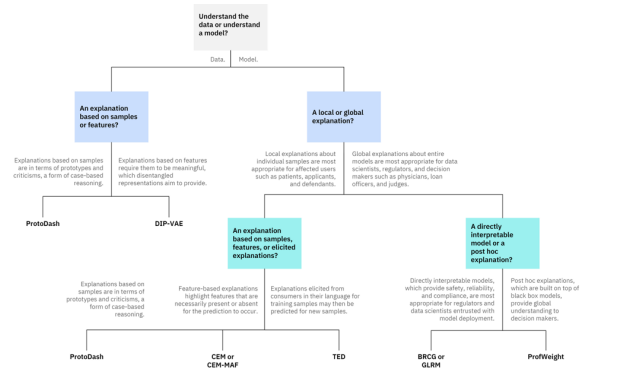


Figure 1: The taxonomy of XAI methods introduced in AIX360

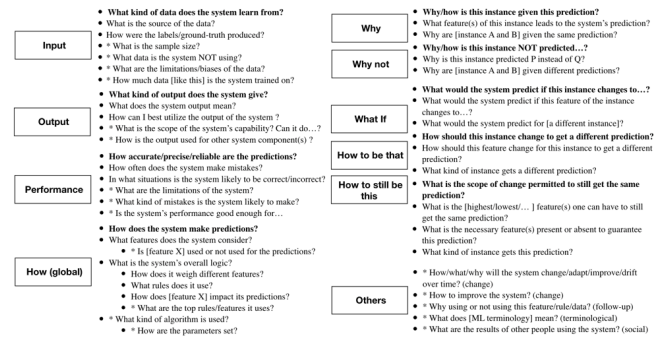


Figure 2: XAI Question Bank as presented in [7]

4.3 Part 3: Introduction to user-centered XAI (15 minutes)

Given the large collection of diverse XAI methods, in design practices it is critical to identify the appropriate method to explain based on application, context and user specific needs. In this part, we will give an overview of existing design guidelines and methods that could support the needs-specification work for XAI and guide a user-centered design process. These include a “question-driven XAI design process” that the course instructor developed by studying the design practices around explainability in various AI product teams [7], utilizing an XAI Question Bank that comprises prototypical questions users have for understanding AI systems (Figure 2)

4.4 Part 4: Hands-on experience (20 minutes)

In the last part, we will walk attendees through some code samples to create explanations that we have discussed in Part 2. Through the experience, attendees could gain a better understanding on what is offered by the open-source toolkits, and how to utilize them for their own work.

5 PRACTICAL WORK

The last part of the hands-on experience will require pre-installation of Python Jupyter Notebook with sample code. Instructions for installation will be sent to the attendees beforehand. This step is

¹<http://aix360.mybluemix.net/resources#guidance>

also optional if an attendee prefers to just watch the instructors demonstrating the code.

Since CHI 2021 will be held in a virtual mode, we will prepare to accommodate remote attendees. All course content should be easily consumable through video conferencing tools. Slides will be sent to attendees prior to the course and made available on course website (<https://hcixaitutorial.github.io>) afterwards.

6 INSTRUCTOR BACKGROUND

Q. Vera Liao is a Research Staff Member in IBM Research AI. Her current research focuses are on explainable AI, human centered machine learning, and conversational agents. Her research work received multiple awards at ACM CHI and IUI, and contributed to IBM's Watson Assistant and AI Explainability 360. She actively organizes events that connect the HCI and AI communities, including several workshops and a panel at CHI, IUI and CSCW. She serves on the Editorial Board of International Journal of Human-Computer Studies (IJHCS) and ACM Transactions on Interactive Intelligent Systems (TiiS), the Organizing Committee for IUI 2019 and CSCW 2021, and as an Associate Chair for multiple CHI and CSCW conferences. She received Ph.D. in Computer Science from the University of Illinois at Urbana-Champaign, and Bachelor in Industrial Engineering from Tsinghua University.

Moninder Singh is a Research Staff Member in the IBM Research AI organization at the IBM T. J. Watson Research Center. He received his Ph.D. in Computer and Information Science from the University of Pennsylvania in 1998. He is primarily interested in developing and deploying solutions for interesting analytics and decision support problems. His main research areas are machine learning and data mining, artificial intelligence, data privacy, information retrieval, probabilistic modeling and reasoning, and text mining. He has been actively working in issues of fairness and trust in AI, has contributed to the IBM's AI Fairness 360 and AI Explainability 360 open source toolkits, and has given several tutorials/talks and published papers on issues relating to trust in AI models.

Yunfeng Zhang is a Research Staff Member at IBM Research AI. His research interests lie in the intersection between HCI and AI. His recent research projects involved creating conversational agents, modeling social interactions, and studying AI explainability, fairness, and trust. He contributed to the IBM's AI Fairness 360 and AI Explainability 360 open source toolkits, which are designed to help AI developers create intelligible and fair AI solutions. He received his Ph.D. in computer and information science from the University of Oregon in 2015.

Rachel K.E. Bellamy is a Principal Research Scientist and Chair of the Computer Sciences Council at IBM T J Watson Research Center. In this role she heads a Council that manages a Research portfolio of exploratory science projects. Prior to this, she led an interdisciplinary team of human-computer interaction researchers, user experience designers and software engineers. That team most recently contributed to several IBM Research's Trusted AI projects, including the AI Fairness 360 and AI Explainability 360. Rachel received her doctorate in cognitive psychology from University of Cambridge, UK. She received a Bachelor of Science in psychology with mathematics and computer science from University of London.

Before coming to IBM Research, she worked in Apple Computer's Advanced Technology Group researching software support for collaborative learning.

7 RESOURCES

Course website: <https://hcixaitutorial.github.io>

AIX 360 toolkit: <http://aix360.mybluemix.net>

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